

M.Tech Semester I Mid Semester Examination,
Session 2024-25 Mining Engineering
Subject: Operations Research in Mining, Subject Code: MN561

Maximum Marks: 30

Time: Two Hour

Answer All question.

Answer All question.		2																																																		
1.	Write detail notes on the assumption of linear programming																																																			
2.	Determine the feasible space for each of the following independent constraints, given that $x_1, x_2 \geq 0$. (a) $-3x_1 + x_2 \leq 6$. (b) $x_1 - 2x_2 \geq 5$. (c) $2x_1 - 3x_2 \leq 12$. (d) $x_1 - x_2 \leq 0$. (e) $-x_1 + x_2 \geq 0$.	5*1																																																		
3.	The following tableau represents a specific simplex iteration. All variables are nonnegative. The tableau is not optimal for either a maximization or a minimization problem. Thus, when a nonbasic variable enters the solution it can either increase or decrease z or leave it unchanged, depending on the parameters of the entering nonbasic variable. <table><tr><th>Basic</th><th>x_1</th><th>x_2</th><th>x_3</th><th>x_4</th><th>x_5</th><th>x_6</th><th>x_7</th><th>x_8</th><th>Solution</th></tr><tr><td>z</td><td>0</td><td>-5</td><td>0</td><td>4</td><td>-1</td><td>-10</td><td>0</td><td>0</td><td>620</td></tr><tr><td>x_8</td><td>0</td><td>3</td><td>0</td><td>-2</td><td>-3</td><td>-1</td><td>5</td><td>1</td><td>12</td></tr><tr><td>x_3</td><td>0</td><td>1</td><td>1</td><td>3</td><td>1</td><td>0</td><td>3</td><td>0</td><td>6</td></tr><tr><td>x_1</td><td>1</td><td>-1</td><td>0</td><td>0</td><td>6</td><td>-4</td><td>0</td><td>0</td><td>0</td></tr></table> (a) Categorize the variables as basic and nonbasic and provide the current values of all the variables.	Basic	x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8	Solution	z	0	-5	0	4	-1	-10	0	0	620	x_8	0	3	0	-2	-3	-1	5	1	12	x_3	0	1	1	3	1	0	3	0	6	x_1	1	-1	0	0	6	-4	0	0	0	2
Basic	x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8	Solution																																											
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x_1	1	-1	0	0	6	-4	0	0	0																																											
4.	The WorldLight Company produces two light fixtures (products 1 and 2) that require both metal frame parts and electrical components. Management wants to determine how many units of each product to produce so as to <u>maximize profit</u> . For each unit of product 1, 1 unit of frame parts and 2 units of electrical components are required. For each unit of product 2, 3 units of frame parts and 2 units of electrical components are required. The company has 200 units of	5																																																		

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	frame parts and 300 units of electrical components. Each unit of product 1 gives a profit of \$1, and each unit of product 2, up to 60 units, gives a profit of \$2. Any excess over 60 units of product 2 brings no profit, so such an excess has been ruled out. (a) Formulate a linear programming model for this problem. (b) Use the graphical method to solve this model. What is the resulting total profit?	
5.	Work through the Simplex method step by step to solve the following problem: Maximize $Z = -x_1 + x_2 + 2x_3,$ subject to $\begin{aligned} x_1 + 2x_2 - x_3 &\leq 20 \\ -2x_1 + 4x_2 + 2x_3 &\leq 60 \\ 2x_1 + 3x_2 + x_3 &\leq 50 \end{aligned}$ and $x_1 \geq 0, \quad x_2 \geq 0, \quad x_3 \geq 0.$	5
6.	MARY has just been diagnosed as having a cancer at a fairly advanced stage. Specifically, she has a large malignant tumor in the bladder area (a "whole bladder lesion"). Mary is to receive the most advanced medical care available to give her every possible chance for survival. This care will include extensive radiation therapy. Radiation therapy involves using an external beam treatment machine to pass ionizing radiation through the patient's body, damaging both cancerous and healthy tissues. Normally, several beams are precisely administered from different angles in a two dimensional plane. Due to attenuation, each beam delivers more radiation to the tissue near the entry point than to the tissue near the exit point. Scatter also causes some delivery of radiation to tissue outside the direct path of the beam. Because tumor cells are typically	3+4+

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microscopically interspersed among healthy cells, the radiation dosage throughout the tumor region must be large enough to kill the malignant cells, which are slightly more radiosensitive, yet small enough to spare the healthy cells. At the same time, the aggregate dose to critical tissues must not exceed established tolerance levels, in order to prevent complications that can be more serious than the disease itself. For the same reason, the total dose to the entire healthy anatomy must be minimized.

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After thorough analysis of this type, the medical team has carefully estimated the data needed to design Mary's treatment, as summarized in Table below. The first column lists the areas of the body that must be considered, and then the next two columns give the fraction of the radiation dose at the entry point for each beam that is absorbed by the respective areas on average. For example, if the dose level at the entry point for beam 1 is 1 kilorad, then an average of 0.4 kilorad will be absorbed by the entire healthy anatomy in the two-dimensional plane, an average of 0.3 kilorad will be absorbed by nearby critical tissues, an average of 0.5 kilorad will be absorbed by the various parts of the tumor, and 0.6 kilorad will be absorbed by the center of the tumor. The last column gives the restrictions on the total dosage from both beams that is absorbed on average by the respective areas of the body. In particular, the average dosage absorption for the healthy anatomy must be as small as possible, the critical tissues must not exceed 2.7 kilorads, the average over the entire tumor must equal 6 kilorads, and the center of the tumor must be at least 6 kilorads.

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TABLE Data for the design of Mary's radiation therapy			
Area	Fraction of Entry Dose Absorbed by Area (Average)		Restriction on Total Average Dosage, Kilorads
	Beam 1	Beam 2	
Healthy anatomy	0.4	0.5	Minimize
Critical tissues	0.3	0.1	≤ 2.7
Tumor region	0.5	0.5	$= 6$
Center of tumor	0.6	0.4	≥ 6

Based on the information given above, formulate this as a linear programming problem. Solve it by Big M method and Two Phase method

$$z^* - 12m = (0.4 - 1.1m)x_1 + (0.5 - 0.9m)x_2 - 0s_1 -$$

$$0.5 - 0.9 = \frac{0.4}{3} + \frac{1.1}{3}m$$

$$-12m - 9 \times 0.4 + 1.1 \times 9m.$$

$$m + \frac{8}{3}m - \frac{11}{6}$$

$$-2.1m - 3.6 + \frac{3}{5} \left(\frac{8m+11}{15} \right)$$